

Bhavesh Nakum

Senior Software Engineer | Backend, Distributed Systems & Big Data | Cloud & AI-Augmented Engineering
Sarjapur Road, Bengaluru, 560035, India | bhaveshnakum.com | linkedin.com/in/bhaveshnakum7 |
bhaveshnakumcoed@gmail.com | +91-8128573526

PROFESSIONAL SUMMARY

Senior Software Engineer with 8+ years of experience designing, developing, and operating high-performance backend systems, distributed microservices, and large-scale data platforms in Java and Python. Deep expertise across event-driven architecture (Apache Kafka, Apache Flink), big data processing (Apache Spark, Apache Airflow, Hive, HDFS), and multi-tenant, multi-region SaaS platform design with a strong focus on high availability, fault tolerance, and secure API development (OAuth, OWASP). Hands-on experience designing AI agent-driven workflows - including an agentic system that generates data schemas from natural-language chat input and a self-service, automated data consumption layer - alongside daily use of AI-augmented development tools (Cursor, Claude Code, GitHub Copilot, Qodo Merge) across coding, design, PR review, and testing. Skilled in cloud-native engineering across Microsoft Azure, Google Cloud Platform, and AWS, with hands-on experience in Kubernetes, Docker, and CI/CD pipeline automation. Proven technical leader with a track record of architecting production systems end-to-end, mentoring engineering teams, participating in technical hiring, and collaborating cross-functionally with product, data science, and business stakeholders across e-commerce, healthcare, and enterprise IT environments.

TECHNICAL SKILLS

Languages: Java, Python, SQL

Backend Development & APIs: Spring Boot, RESTful Web Services, Microservices, Backend-for-Frontend (BFF), Distributed Systems, Event-Driven Architecture, Java Concurrency (Threads, Executors, CompletableFuture, Reactive Patterns)

Security: OAuth, OWASP Secure Coding Practices

AI Agents & Automation: Agentic Workflow Design, Automated Schema Generation from Natural Language, Self-Service Data Pipeline Automation

Big Data & Streaming: Apache Kafka, Apache Spark, Apache Flink, Apache Airflow, Hive, HDFS, Sqoop

Data Modeling & Architecture: Schema Design, Key Strategy & Partitioning, Grain Definition, Multi-Tenant Data Isolation, NoSQL Data Modeling

Cloud Platforms: Microsoft Azure (Data Factory, Databricks, Synapse, Key Vault), Google Cloud Platform (DataProc, BigQuery, GCS), AWS (Lambda, S3, SQS)

High Availability & Resilience: Multi-Region Replication, High Availability, Fault Tolerance, Blue-Green & Canary Deployments, Autoscaling

Databases: PostgreSQL, MySQL, Azure Cosmos DB, Cassandra (Relational & NoSQL)

CI/CD & DevOps: Jenkins, Docker, Kubernetes, Helm, Maven, Git, CI/CD Pipeline Automation, Infrastructure Automation, Agile/Scrum/SDLC

Testing & Quality: JUnit, Mockito, Cucumber, Test-Driven Development (TDD), Unit/Functional/E2E Testing

Observability & Monitoring: Grafana, Micrometer, Custom Metrics & Alerting, Monitoring & Dashboarding

AI-Augmented Engineering: Cursor, Claude Code, GitHub Copilot, Qodo Merge, Claude, ChatGPT (applied to coding, design, PR review, testing, and research)

Familiarity (exposure, non-production depth): Snowflake, GraphQL, Hibernate/JPA, Elasticsearch

PROFESSIONAL EXPERIENCE

Senior Software Engineer

eBay, Bengaluru | June 2026 - Present

- **Data Platform Ownership:** Member of eBay's core Data team, owning end-to-end design, development, and maintenance of real-time and batch data pipelines across the eBay platform; responsible for reliable delivery of eBay Live feature data to downstream data consumers using Apache Flink, Apache Kafka, SQL, and Hadoop.
- **AI Agent-Driven Schema Onboarding:** Designed and implemented an AI agent workflow within the team's data consumption layer that automatically generates data schemas from natural-language chat input and user-provided data, replacing manual schema-definition forms and streamlining the onboarding of new datasets.
- **Self-Service Automated Data Consumption:** Built an automated data consumption layer enabling downstream consumers to write and run their own SQL queries to extract data on a batch, scheduled, or on-demand basis - eliminating the need for the data management platform team to manually author extraction queries and pipelines for each consumer request.
- **AI-Augmented Development:** Use Cursor and Claude Code for coding, PR reviews, design, and testing as part of daily engineering practice.

Senior Software Engineer (Lead)

Walmart, Bengaluru | Apr 2025 - May 2026

- **Distributed, Event-Driven Architecture:** Architected Java and Spring Boot microservices on a Kafka-based event-driven architecture, processing 5K+ events per second with sub-second latency to power dynamic repricing across millions of SKUs.
- **Multi-Tenant SaaS Platform Design:** Designed the schema and key strategy for a multi-market pricing platform spanning Canada, Mexico, Chile, and additional markets, with configurable tenant-level isolation supporting both database-level separation and tenant-based table partitioning to meet country-specific data compliance requirements.
- **High Availability Architecture:** Deployed database and application infrastructure across 2 active regions and 1 replica region to eliminate single-region outage risk and optimize regional availability and latency.
- **CI/CD, Containerization & Deployment:** Built Docker and Kubernetes-based deployment templates (including Helm) within Walmart's internal release pipeline, packaging code into deployable artifacts and applying blue-green deployment strategy for Kafka-based services and canary deployment for API-based applications, with automated test execution, post-deployment code coverage checks, metrics exposure, and autoscaling-based alerting.
- **Test-Driven Development & Code Quality:** Practiced Test-Driven Development, writing tests before implementation with automated coverage via JUnit, Mockito, and Cucumber; led a structured PR review and merge process for a team of 6 engineers, enforcing design doc adherence, the KISS principle, and a 60/40 positive-to-negative test case ratio.
- **Secure, Well-Architected APIs:** Designed, deployed, and maintained RESTful APIs and Backend-for-Frontend (BFF) services applying OAuth-based authentication and OWASP secure coding practices, tailored to different client application needs.
- **Observability & Cost Optimization:** Designed and delivered a reusable, plug-and-play Micrometer metrics module adopted across engineering teams for self-service observability and custom Grafana alerting; reduced infrastructure cost and improved availability by fine-tuning CPU/memory allocation, enabling autoscaling, and consolidating deployables.
- **AI-Augmented Development:** Used GitHub Copilot for development work and design assistance, and Qodo Merge for automated PR review on GitHub, integrated into daily engineering workflow.
- **Technical Leadership & Collaboration:** Led and mentored a cross-functional team of 6 engineers, participated in hiring and interview panels for vendor and FTE roles, and partnered quarterly with product managers and data scientists to define feature roadmaps and plan cross-functional releases.

Software Developer

Walmart, Bengaluru | Apr 2022 - Mar 2025

- **Pipeline Engineering:** Built real-time anomaly detection and repricing pipelines, removing 1M+ mispriced items from the Walmart catalog.

- **Seller Tools:** Developed pricing automation tools for sellers, reducing manual overhead and improving competitiveness.
- **Reliability:** Enhanced system reliability with full-stack alerting and dashboarding for production-grade observability.

Data Engineer

UnitedHealth Group, Hyderabad | Apr 2021 - Apr 2022

- **ETL Design:** Designed scalable ETL pipelines using Azure (Data Factory, Databricks, Synapse) to ingest and transform healthcare data across supported asset classes and data domains.
- **Automation:** Developed PySpark-based transformation logic and modular Airflow DAGs to accelerate pipeline development while ensuring HIPAA compliance.

Systems Engineer

Tata Consultancy Services, Hyderabad | Sep 2018 - Apr 2021

- **Cloud Migration:** Migrated legacy data warehousing platforms to a Hadoop-based Azure ecosystem for Proximus (Belgium).
- **Optimization:** Rewrote HQL queries to optimize performance and reduce compute costs by 25%.
- **Modernization Tools:** Engineered a Python-based tool to convert Teradata SQL to Hive and PySpark scripts using Python and Regex for Bank of America.

ADDITIONAL HIGHLIGHTS

- **Agentic AI System Design:** Hands-on experience designing and shipping an AI agent-driven workflow for automated data schema generation and a self-service, automated data consumption layer - moving a previously manual, team-dependent process to a fully automated, consumer-driven path.
- **Mentorship & Hiring:** Mentored new hires and junior engineers; participated in hiring and interview panels for vendor and FTE roles.
- **AI-Augmented Development:** Practical, day-to-day use of AI coding tools (Cursor, Claude Code, GitHub Copilot, Qodo Merge) and general-purpose LLM tools (Claude, ChatGPT) to accelerate development, elevate code review quality, and support research and technical communication.
- **Legacy & Proprietary System Modernization:** Consistent track record diving into proprietary and legacy systems - internal release pipelines, Teradata data warehouses, Hadoop platforms - to understand their inner workings and strategically modernize them.
- **Regulated & Enterprise Environments:** Experience delivering technology solutions in compliance-driven settings (HIPAA-regulated healthcare data) and for large enterprise clients (Proximus/Belgium, Bank of America).
- **CI/CD:** Automated CI/CD pipelines by integrating functional, integration, end-to-end, and performance testing into the deployment process.
- **Engineering Standards:** Passionate about performance optimization, code quality, testability, and technical documentation for high-concurrency, high-throughput production systems.

EDUCATION & CERTIFICATIONS

- B.Tech - Computer Engineering: SVNIT, Surat | 2018 | CGPA: 7.18 / 10
- Microsoft Certified: Azure Fundamentals
- Apache Airflow Fundamentals